1. Introduction

The corpus callosum is the major white-matter structure connecting the two cerebral hemispheres. It is commonly viewed as the main route for sharing sensory information and synchronizing processing across the hemispheres . Corpus callosum dysgenesis (CCD), occurring in roughly 1 in 4,000 live births , refers to a range of developmental anomalies in this structure, from partial underdevelopment to complete absence. No single gene or combination of genes has been identified across patients, and the condition does not appear to be strongly inherited (Lau, 2012). Individuals with CCD exhibit highly heterogeneous clinical profiles, ranging from asymptomatic presentations to cases involving psychomotor and communication difficulties, epilepsy, and frequent comorbidities with DSM-5 neurodevelopmental conditions, including traits resembling those observed in autism spectrum disorder (ASD) . Interestingly, most people with CCD do not display the classic split-brain syndrome, in which the two hemispheres cannot communicate, suggesting that alternative pathways or compensatory strategies can maintain interhemispheric coordination. Studying how connectivity in the absence of a corpus callosum relates to autistic-like symptoms can provide key insights into the role of this structure and how the brain adapts and reorganizes when a major interhemispheric pathway is missing.

2. Corpus callosum dysgenesis: development, phenotype, and connectivity

2.1 The role of the CC

The corpus callosum (CC) plays a central role in interhemispheric coordination by regulating how information and processing resources are distributed between the two cerebral hemispheres (Banich, 1998). At a basic level, it enables the transfer of sensory and perceptual information across hemispheres, ensuring that inputs from one side of the body or visual field are accessible to the contralateral cortex (Banich, 1998; Gazzaniga, 2000). However, its function extends far beyond simple information transfer. The CC contributes to the organization of cognitive processing by dynamically balancing hemispheric specialization and interhemispheric integration. Regarding specialization, functional lateralization improves efficiency by allowing one hemisphere to specialize in specific operations, reducing redundancy and limiting interhemispheric conduction delays. The CC supports this specialization by regulating interhemispheric interference: callosal projections terminate in cortical layers where they can engage both excitatory pyramidal neurons and inhibitory GABAergic interneurons, allowing activity in one hemisphere to suppress competing activity in the other when necessary. This mechanism enables stable lateralization while preserving flexibility.

At the same time, the CC enables the coordination of processing across hemispheres when task demands exceed the capacity of a single hemisphere, which requires interhemispheric integration. It has been proposed that the CC acts as a dynamic attentional module, allowing one hemisphere to recruit processing resources from the other, thereby supporting parallel computation and increasing overall processing capacity (Banich, 1998). Neuropsychological and functional studies support this hypothesis by showing that although each hemisphere can perform simple perceptual or motor operations independently, complex cognitive functionsâ€"such as decision-making, contextual integration, or multi-domain processingâ€"require efficient interhemispheric communication and precise temporal coordination (Gazzaniga, 2000; Hinkley et al., 2016; Yuan et al., 2020). Thus, rather than enforcing constant coupling between hemispheres, the CC acts as a regulatory structure that flexibly modulates interhemispheric connectivity, ensuring an optimal trade-off between specialization, parallel processing, and integration.

2.2 The emergence of DCC

The corpus callosum develops through a tightly regulated sequence during early brain ontogeny. Cortical projection neurons extend axons toward the midline, where specialized glial structuresâ€"such as the glial wedge, midline zipper glia, and indusium griseumâ€"provide guidance cues that enable axons to cross to the contralateral hemisphere and innervate homotopic targets (Tovar-Moll et al., 2006; Paul et al., 2012). Disruptions in these guidance mechanisms prevent callosal axons from crossing, resulting in partial or complete dysgenesis of the corpus callosum. When axons fail to cross, they usually form longitudinal Probst bundles, which remain ipsilateral and are consistently observed in humans with DCC (Tovar-Moll et al., 2006; Lazarev et al., 2016). Some individuals also develop sigmoid bundles, connecting non-homologous regions (Tovar-Moll et al., 2006). Despite this profound disruption of the principal interhemispheric pathway, individuals with DCC do not exhibit the classical split-brain syndrome observed after surgical callosotomy in adulthood, which suggests that functional integration across hemispheres is preserved through alternative developmental trajectories. Rather than producing a simple loss of interhemispheric communication, early callosal maldevelopment appears to trigger large-scale reorganization of neural circuitry, leading to a distinct but viable configuration of brain connectivity.

2.3 Consequences of corpus callosum absence on brain circuitry

Despite the absence of the corpus callosum, resting-state fMRI studies show that the qualitative organization of large-scale resting-state networks is largely preserved in individuals with dysgenesis of the corpus callosum. Owen et al.Â (2013) reported that canonical resting-state networks identified in neurotypical controls can also be reliably detected in DCC, indicating that the global functional architecture of the brain can emerge even without direct callosal connections. Neuroimaging evidence indicates that this apparent preservation is accompanied by profound network reorganization, characterized by reduced interhemispheric coupling and a shift toward increased intrahemispheric connectivity. In particular, functional connectivity studies have shown enhanced within-hemisphere integration and stronger local clustering in DCC, alongside weakened homotopic interhemispheric synchronization (Hinkley et al., 2012). Graph-theoretical analyses further support this pattern, revealing markedly reduced interhemispheric functional connectivity together with increased local and intrahemispheric structural connectivity (Yuan et al., 2020; Shi et al., 2021).

Such a shift toward within-hemisphere processing has led to the hypothesis that information may become bilaterally represented within each hemisphere, thereby reducing the need for direct interhemispheric transfer. If this were the case, functions that typically require callosal communication should remain intact because each hemisphere would contain redundant representations. For example, under typical organization, tactile input from the left hand is processed in the right somatosensory cortex and must be transferred via the corpus callosum to left-hemisphere language areas for verbal identification; true bilateral representation would therefore predict preserved left-hand naming even in the absence of callosal transfer. Monteiro et al.Â (2019) directly tested this prediction in the cheirosensory domain. Although individuals with callosal dysgenesis showed intact left-hand naming, neural responses remained strongly lateralized, with no evidence of ipsilateral somatosensory activation. Thus, preserved performance did not reflect genuine bilateral duplication of function. Instead, the persistence of hemispheric specialization suggests compensatory mechanisms, such as indirect interhemispheric pathways or alternative functional reorganization that bypass canonical callosal transfer. These findings point toward atypical structural reconfiguration rather than true bilaterality, motivating closer examination of the anatomical routes that support interhemispheric integration in DCC.

In DCC, callosal axons that fail to cross the midline instead form ipsilateral longitudinal bundles known as Probst bundles, that preserve the topographic organization normally observed in callosal fibers (Tovar-Moll et al., 2006). Rather than supporting interhemispheric communication, this preserved organization likely reflects a developmental reallocation of callosal axons toward intrahemispheric circuitry, potentially reinforcing within-hemisphere integration. This interpretation is consistent with converging structural and functional evidence for increased intrahemispheric connectivity and reduced interhemispheric integration in individuals with callosal dysgenesis (Paul et al., 2012; Yuan et al., 2020; Shi et al., 2021). Additionally, sigmoid bundles are aberrant interhemispheric fiber tracts observed in some DCC cases that curve around the ventricles to connect non-homologous cortical regions across hemispheres. Their preferential occurrence in the most severe forms of callosal malformation suggests maladaptive miswiring rather than a compensatory mechanism (Owen et al., 2013).

Beyond intrahemispheric reorganization, alternative commissural pathways may contribute to residual interhemispheric transfer in DCC. The anterior commissure, which normally connects anterior temporal regions, is frequently enlarged in DCC (Hetts et al., 2006), and its microstructural integrity has been associated with better cognitive and attentional performance (Siffredi et al., 2019), suggesting partial functional compensation. The posterior commissure does not typically support large-scale cortical integration, but it provides a dorsal midbrain crossing point that may serve as an alternative conduit in the absence of the corpus callosum. Consistent with this view, Tovar-Moll et al.Â (2014) identified anomalous interhemispheric bundles connecting parietal regions through the midbrain and ventral forebrain, crossing via the anterior or posterior commissure. These findings therefore indicate structural rerouting rather than simple amplification of pre-existing functions. However, transmission through these alternative pathways is necessarily slower and less temporally precise than through callosal fibers, as they often involve longer and potentially polysynaptic routes with fewer fast, densely myelinated cortico-cortical fibers. This is consistent with electrophysiological and neuroimaging evidence of delayed interhemispheric transfer and reduced synchronization in DCC, particularly for higher-order associative networks (Paul et al., 2012; Hinkley et al., 2016).

An alternative, but not mutually exclusive, interpretation is that subcortical circuits play a central role in preserving bilateral functional connectivity, particularly within sensory systems. In the healthy brain, thalamocortical projections relay sensory information via largely symmetrical bilateral connections to primary sensory cortices. In DCC, resting-state fMRI shows that while interhemispheric connectivity is reduced for associative hubs, it is relatively preserved for primary sensory networks, consistent with compensation via subcortical routes that help enforce bilaterally coordinated activity (Owen et al., 2013).

Overall, all these findings support the hypothesis that, in the absence of the corpus callosum, the brain recruits a heterogeneous set of structural and subcortical adaptations to preserve interhemispheric connectivity, with the relative contribution of each pathway varying across individuals due to developmental plasticity and anatomical constraints. This reorganization provides an essential framework for interpreting the cognitive and behavioral characteristics associated with dysgenesis of the corpus callosum.

2.4 Consequences of corpus callosum absence on behavior

According to Brown (2019), the core syndrome of dysgenesis of the corpus callosum (DCC) includes three main aspects: (1) reduced interhemispheric transfer of sensory-motor information, (2) slowed cognitive processing, and (3) deficits in complex reasoning and novel problem-solving. First, individuals with DCC show reduced interhemispheric transfer of sensory-motor and tactile-spatial information. Experimental paradigms requiring cross-hemispheric integrationâ€”such as tasks in which tactile input from one hand must be verbally identifiedâ€—consistently reveal increased interhemispheric transfer time and reduced efficiency compared to neurotypical controls (Brown et al., 1999; Paul et al., 2007). Transfer is generally sufficient for simple unimanual tasks but becomes less efficient when precise bilateral coordination or integration of lateralized sensory input is required, reflecting the role of the corpus callosum in synchronizing activity across hemispheres. Developmentally, the gradual maturation of the corpus callosum parallels improvements in bimanual coordination, underscoring its contribution to the acquisition of coordinated motor and tactile-spatial skills (Mueller et al., 2009; Gooijers et al., 2013). Second, individuals with DCC exhibit slowed cognitive processing, which has been identified as a primary contributor to executive control difficulties (Brown, 2019; Marco et al., 2012; Mathias et al., 2004). This slowing is typically reflected in increased reaction times across a range of cognitive tasks, even when accuracy is preserved. It appears to affect processing speed broadly rather than a single domain, and may limit the efficient coordination of multiple cognitive operations, thereby impacting executive functions such as inhibition, cognitive flexibility, and working memory. Third, individuals with DCC show impairments in complex reasoning and novel problem-solving. Tasks requiring abstraction, flexible strategy formation, generalization, or integration of multiple conceptual elements are particularly affected, whereas overlearned or routine tasks remain relatively preserved (Brown, 2019; Koivisto, 2000; Schieffer, 2000; Erickson, 2013; Cherbuin, 2005). This profile highlights the importance of interhemispheric communication for high-level cognitive integration beyond basic sensory-motor coordination.

As a consequence of these primary deficits, individuals with DCC often show difficulties in social and emotional functioning. They may have trouble recognizing complex emotions, understanding social cues, and integrating cognitive and emotional information in social contexts (Paul, 2021; Bridgman, 2014; Anderson, 2017). For example, while basic emotional experience is typically intact, the ability to interpret nuanced emotional expressions or predict othersâ€™ behavior is impaired, reflecting deficits in higher-order social cognition rather than in perception (Paul, 2021; Kang, 2009). Consistent with this distinction, Symington et al.Â (2010) reported impairments in theory of mind and in the interpretation of complex social intentions, whereas more basic aspects of social perception were relatively preserved. Difficulties also extend to adaptive social behavior: individuals may apply social rules rigidly, struggle with flexible decision-making, or fail to use context to guide responses (Brown, 2021; Mangum, 2021; Marco, 2012). These impairments are particularly pronounced in complex or novel situations, where effective interhemispheric communication would normally support the integration of multiple informational streams, and are often accompanied by reduced metacognitive awareness of oneâ€™s own difficulties (Mangum, 2018, 2021; Kaplan, 2012).

These social and cognitive characteristics overlap with traits commonly described in autism spectrum conditions, including difficulties in social reciprocity, rigid behavioral patterns, and reduced use of contextual information. Although DCC and autism are distinct conditions, this convergence raises the possibility that alterations in large-scale neural connectivityâ€—central in DCC and widely discussed in autism researchâ€—may contribute to partially similar behavioral profiles. To examine this potential convergence more closely, we next consider the behavioral phenotype of autism and the neural mechanisms proposed to underlie it.

3 Description of autism

3.1 Behavioral description

Autism, or Autism Spectrum Disorder (ASD), is a neurodevelopmental condition affecting about 3% of the general population . Autistic phenotypes are highly heterogeneous, ranging from severe presentations to more subtle forms, sometimes described as the broad autism phenotype . There is no biological test or neuroimaging marker to confirm the diagnosis; it is determined entirely through behavioral observation. Twin studies indicate a strong genetic contribution , although no single gene is sufficient to account for the condition, suggesting complex and multifactorial mechanisms. According to the Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition , the diagnosis is based on two core criteria: (A) persistent deficits in social communication and social interaction across contexts, and (B) restricted and repetitive patterns of behavior, interests, or activities. Criterion A includes difficulties in social-emotional reciprocity, impaired understanding and use of social cues (such as eye contact, facial expressions, gestures, and prosody), and challenges in developing, maintaining, and understanding relationships. Criterion B encompasses repetitive motor movements, insistence on sameness and inflexible adherence to routines, highly restricted or fixated interests, and hyper- or hypo-reactivity to sensory input. Symptoms must be present early in development and cause clinically significant impairment. In addition to the core diagnostic domains, a range of associated features are consistently reported in ASD, including motor coordination difficulties , executive dysfunction , and elevated anxiety . Additionally, attentional processes appear atypical in ASD, in particular with respect to impaired attentional disengagement , reduced cognitive flexibility and shifting , and altered allocation of attention to socially salient stimuli .

Although not included in the formal diagnostic criteria, these dimensions contribute substantially to the broader autistic phenotype.

Several classical cognitive theories of autism, primarily developed in the 1980s and 1990s, have attempted to account for the diverse manifestations of autism within a single explanatory framework. Executive function theories attribute autistic symptoms to core difficulties in cognitive control, such as planning, flexibility, and inhibition , while social cognition theories focus on difficulties in understanding othersâ€™ emotions and mental states, reflected in altered processing of socially relevant stimuli such as faces or voices . The Weak central coherence account emphasizes a general bias toward local over global processing, both at the perceptual and conceptual level (focusing on details rather than a meaningful whole), leading in particular to reduced integration of contextual information . However, accumulating evidence indicates that the putative core deficit proposed by each account is neither systematic nor specific. For instance, a detail-focused or local processing bias is not consistently observed across autistic individuals and can coexist with intact global processing . Similarly, executive function impairments, often invoked to explain rigidity and repetitive behavior, are also prominent in other neurodevelopmental conditions such as ADHD, in which social deficits are not a defining feature . These observations challenge single-deficit models and suggest that no isolated cognitive mechanism fully captures the breadth and heterogeneity of the autistic phenotype.

In response to these limitations, more recent accounts have proposed that autistic characteristics arise from the interaction of multiple non-specific cognitive constraints, which become apparent primarily under conditions of high informational and integrative demands . In this view, difficulties emerge most clearly when tasks require the integration of multiple cues into higher-order representations , rather than during isolated executive, social, or perceptual operations. Empirical findings support this perspective: studies report largely preserved accuracy for basic social, perceptual, and contextual processing in ASD, accompanied by slower responses, but selective impairments when task complexity is maximalâ€”particularly when successful performance depends on the integration of global information or complex social signals {ben-yosef2016}. Importantly, this complexity-based account does not posit a single primary cognitive deficit; rather, it characterizes autism as a condition in which integrative processing becomes constrained under increasing computational demands.

Recent alternative, not necessarily incompatible accounts have increasingly proposed that sensory hypersensitivity, together with consequential alterations in attentional regulation, may constitute a primary organizing dimension of autism, rather than secondary features (Ornitz, 1988; Kinsbourne, 1991; Liss et al., 2006). Within this perspective, chronic sensory overreactivity is thought to bias the system toward overselective attention as a compensatory mechanism aimed at managing heightened arousal and reducing overwhelming input, characterized by intense focus on a restricted subset of stimuli and difficulty disengaging or shifting (Liss et al., 2006; Oâ€™Riordan et al., 2001). Such narrowing of attentional scope can in turn promote perseverative and repetitive behaviors, which may function as regulatory strategies that reduce environmental unpredictability and stabilize arousal levels (Kinsbourne, 1991; Liss et al., 2006). Because social interaction requires rapid, flexible coordination of multiple dynamic cues (e.g., gaze, facial expression, prosody, contextual information), an overfocused and inflexible attentional style may also disproportionately affect social communication. In this framework, sensory hypersensitivity, attentional rigidity, repetitive behavior, and social-communicative difficulties are conceptualized as interrelated consequences of a common alteration in arousal and attentional allocation, suggesting that this sensoryâ€“attentional dimension may represent a more proximal mechanism than domain-specific social or executive deficits.

3.1 Neuro correlates of autism

Both the complexity account and the sensoryâ€“attentional account are coherent with converging neurobiological findings. Structural and functional neuroimaging studies report a coordinated pattern of heightened local connectivity alongside reduced long-distance integration, rather than a simple global disconnection (Just et al., 2004; Belmonte et al., 2004). Sensory and perceptual cortices appear to be locally overconnected, producing broadly amplified and less selective neural responses that more readily exceed activation thresholds. Such reduced selectivity blurs the distinction between target and competing inputs, effectively lowering signal-to-noise discrimination at early processing stages. As a result, downstream associative regions may receive degraded or poorly structured input and show reduced or inefficient activationâ€”not necessarily due to intrinsic dysfunction, but because of atypical early-stage filtering (Belmonte et al., 2004). These connectivity patterns are consistent with models of altered excitationâ€“inhibition balance and increased neural noise (Hussman, 2001), as well as post-mortem observations of atypical cortical minicolumn organization suggestive of reduced inhibitory surround and enhanced local coupling (Casanova et al., 2002).

This connectivity profile is consistently observed across imaging modalities. Structural MRI studies report region-specific cortical enlargement, particularly in parietal and sensory regions, suggesting disproportionate local cortical development (Piven et al., 1997; Courchesne et al., 2001). Diffusion MRI studies reveal an overrepresentation of short-range fibers alongside reduced integrity of long-range white-matter tracts, including callosal pathways, reflected in lower fractional anisotropy and increased radial diffusivity in a subset of autistic individuals (Alexander et al., 2007; Shukla et al., 2010; Casanova et al., 2002). Functional MRI findings further support this pattern, showing preserved or enhanced local synchronization within regions but reduced functional coupling between distant nodes of large-scale networks involved in executive, social, and integrative processing (Just et al., 2006; Mason et al., 2008). Together, these converging results point to a connectivity profile dominated by strong local coupling and weakened large-scale integration, with direct implications for autistic behavioral phenotypes.

A plausible neurodevelopmental origin of altered connectivity in autism involves early disruptions of inhibitory signaling mediated by Î³-aminobutyric acid (GABA), the brainâ€™s main inhibitory neurotransmitter (Rubenstein & Merzenich, 2003; Belmonte et al., 2004). During brain development, GABAergic neurons limit how strongly and how broadly neurons activate together, thereby shaping activity-dependent synaptic selection and elimination (Hensch, 2005; Ben-Ari, 2002). In the cerebellum, this inhibitory role is primarily carried by Purkinje cells, which are GABAergic neurons and provide the sole output of the cerebellar cortex. Post-mortem and imaging studies report reduced cerebellar volume and a selective loss of Purkinje cells in autism (Courchesne et al., 1994; Fatemi et al., 2012). Reduced inhibitory regulation during early development allows broader and less specific patterns of neural co-activation, increasing the likelihood that weak or poorly specified synaptic connections are retained. This excessive stabilization of synaptic connections is expected to increase neural tissue volume, providing a mechanistic account of early brain overgrowth observed in autism (Courchesne et al., 2001; Hazlett et al., 2005). As development continues, synaptic elimination operates on this inefficiently organized network, yielding atypical pruning and persistent alterations in large-scale functional connectivity.

Together, these findings suggest that altered excitationâ€“inhibition balanceâ€”evidenced by GABAergic disruption, Purkinje cell abnormalities, and a pattern of increased short-range alongside reduced long-range connectivityâ€”offers a unified framework linking sensory, attentional, and integrative accounts of autism. Locally amplified and less selective processing provides a plausible basis for sensory hypersensitivity and heightened reactivity, while reduced long-distance coordination constrains the integration of distributed information, particularly under conditions requiring complex, multi-cue processing. Within this architecture, behavioral rigidity and restricted interests can be understood as stabilizing responses to intensified and poorly filtered input, and social-communicative difficulties as emerging when large-scale coordination is required to flexibly combine perceptual, contextual, and affective signals.

Comparison between ASD and DCC

Given the substantial overlap in clinical features between autism and DCC, it is likely that some aspects of altered connectivity are shared. Although autism does not involve a complete absence of the corpus callosum, both conditions are marked by disruptions in large-scale integration: DCC through the absence or malformation of corpus callosum, and autism through reduced long-range connectivity. In both cases, tasks requiring integration of distributed informationâ€”particularly across hemispheres or distant cortical regionsâ€—appear especially vulnerable. Moreover, while autism has been described in terms of increased short-range or local overconnectivity, DCC is likewise associated with atypical strengthening of alternative intrahemispheric or subcortical pathways that may partially compensate for callosal absence, potentially resulting in a comparable imbalance between local and global connectivity. Despite distinct developmental origins, partially convergent patterns of network organization may therefore contribute to similar cognitive and behavioral outcomes, highlighting the need for direct comparative investigation.

Behavioral comparison of ASD and DCC : Autistic symptoms in DCC

Across comparative studies, individuals with DCC show elevations in social interaction difficulties, theory of mind, emotion understanding, and pragmatic communication relative to neurotypical controls, though typically to a lesser extent than individuals with ASD (Badaruddin et al., 2007; Lau et al., 2012). Executive and attentional differences follow a similar pattern: impairments in cognitive flexibility, sustained attention, and susceptibility to interference are consistently observed in ASD and are also present in DCC, but generally at an intermediate level between controls and autistic participants (Badaruddin et al., 2007; Paul et al., 2014). Notably, measures tapping socially embedded attentional demandsâ€”such as rapid integration of contextual or prosodic cuesâ€—tend to accentuate this gradient pattern. Across domains, DCC participants often occupy a position between neurotypical and ASD groups, suggesting partial but not complete phenotypic overlap. Moreover, social difficulties in DCC may emerge later in development or become more apparent in complex real-world contexts, which may contribute to under-recognition of overlapping autistic features (Booth et al., 2009).

The clearest divergence between DCC and ASD emerges in studies that directly compare the RRBI domain. Whereas ASD is characterized by circumscribed interests, stereotyped behaviors, insistence on sameness, and atypical sensory responsivity, these features are generally less pronounced in DCC. In comparative samples, repetitiveâ€“restrictive behaviors tend to be the least affected domain in DCC relative to ASD (Badaruddin et al., 2007), with lower frequencies of stereotyped movements and intense restricted interests reported on standardized measures (Pazienza et al., 2010; Lau et al., 2012). Although some individuals with DCC may exhibit preference for routine or mild perseverative tendenciesâ€”often discussed in relation to executive functioningâ€—these behaviors are typically less pervasive and less central to the behavioral profile than in ASD. Evidence for a detail-focused perceptual style is likewise limited in DCC cohorts (Booth et al., 2010). With respect to sensory features, most early comparative studies did not report robust elevations in hypersensitivity in DCC; however, more recent work has suggested that atypical sensory responsivity may be present in a subset of individuals with callosal anomalies, albeit with a different profile and lower severity than in ASD (e.g., Demopoulos et al., 2015). Overall, direct behavioral comparisons indicate a relative attenuation of RRBI and sensory-driven features in DCC compared to ASD, even when social-communicative differences are clearly observed.

Taken together, DCC mirrors a substantial portion of the social-cognitive and executive components of ASD but does not recapitulate the full triadic phenotype. Its profile is both quantitatively milder and qualitatively incomplete, supporting the view that the autistic triad reflects separable dimensionsâ€”only some of which appear to depend strongly on interhemispheric integration. One parsimonious hypothesis, consistent with current behavioral and neurobiological data, is that the overlap between ASD and DCC may be driven primarily by vulnerabilities in long-range connectivity and large-scale integration. In DCC, the absence or malformation of the corpus callosum directly constrains interhemispheric communication; in ASD, converging neuroimaging studies report reduced long-distance coordination and inefficient cross-network integration (e.g., Just et al., 2004; Belmonte et al., 2004). These shared integrative constraints could plausibly account for convergent difficulties in complex social cognition, contextual processing, and executive control. By contrast, features more strongly associated with local circuitry alterationsâ€”such as sensory hypersensitivity, heightened reactivity, and pronounced repetitiveâ€“restrictive behaviorsâ€”appear attenuated or inconsistently expressed in DCC, and have been more closely linked in ASD to increased short-range connectivity and excitationâ€“inhibition imbalance (Casanova et al., 2002; Hussman, 2001). This pattern suggests a tentative dissociation whereby large-scale integrative disruption may underlie the shared social-cognitive phenotype, whereas additional local circuit alterations may contribute to the full expression of RRBI and sensory hyperreactivity in ASD. However, this interpretation remains inferential: only direct comparative studies combining fine-grained connectivity measures with dimensional symptom assessments across both conditions will be able to determine whether overlapping behavioral features indeed map onto shared network-level mechanisms.

An additional and theoretically important observation concerns the dissociation between structural extent and behavioral severity. Despite representing a more overt anatomical alteration than the callosal reductions typically reported in ASD (Frazier & Hardan, 2009; Paul et al., 2007), complete agenesis of the corpus callosum is often associated with a milder behavioral phenotype. Moreover, within DCC, complete agenesis does not systematically result in more severe social-communicative impairments than partial dysgenesis, a pattern attributed to early developmental reorganization and compensatory recruitment of alternative pathways (Tovar-Moll et al., 2014; Brown, 2019). This discrepancy suggests that behavioral outcomes cannot be inferred from structural disruption alone and instead depend on how large-scale networks reorganize over development. Direct comparisons of structural and functional connectivity across ASD and DCC are therefore essential to clarify how different patterns of network organization relate to overlappingâ€”or divergentâ€”clinical profiles.

Abnormal corpus callosum in ASD

Neuroimaging comparisons are considerably more limited than behavioral ones. Directly contrasting the corpus callosum in ASD and DCC is obviously not an option, however, it remains theoretically informative to examine whether the corpus callosum in autism shows structural or microstructural particularities that could contribute to altered large-scale integration. In fact, alterations of the corpus callosum (CC) provide a structural substrate for the broader pattern of enhanced local processing combined with reduced long-range integration. Early structural MRI studies reported reduced callosal sizeâ€”particularly when controlling for overall brain volumeâ€”alongside relatively enlarged cortical regions, especially in parietal areas (Piven et al., 1997). Subsequent diffusion imaging studies have consistently identified reductions in callosal microstructural integrity, most prominently in the genu and splenium, reflected in decreased fractional anisotropy, reduced fiber density, or altered fixel-based metrics suggestive of diminished axonal density (Just et al., 2006; Booth et al., 2011; Frazier & Hardan, 2009; Dimond et al., 2019). These alterations disproportionately affect anterior and posterior segments supporting fronto-parietal and temporo-parietal interactionsâ€”networks critically involved in executive control, theory of mind, and multimodal integration.

Functional imaging further indicates that, although activation within individual cortical regions is often preserved, connectivity between distributed regionsâ€—particularly across hemispheresâ€—is reduced (Just et al., 2006). Reduced callosal integrity is associated with decreased interhemispheric functional connectivity and correlates with behavioral measures of social impairment; for example, lower splenial fiber density has been shown to predict greater social symptom severity (Dimond et al., 2019). Anatomical evidence of a higher proportion of short-range relative to long-range fibers (Casanova et al., 2002) suggest that CC alterations form the interhemispheric expression of a broader bias toward local over distributed connectivity. Importantly, these callosal differences are not uniform: group-level reductions coexist with substantial overlap between autistic and control individuals, and milder but more generalized white matter alterations beyond the CC (Dimond et al., 2019), indicating that interhemispheric under-connectivity represents a graded and heterogeneous mechanism contributing to symptom variability rather than a categorical structural deficit (Paul et al., 2007; Frazier & Hardan, 2009).

Lin between connectivity and symptomatology

Most direct comparative studies to date, in addition to rarely including neuroimaging data, have primarily focused on contrasting the prevalence or severity of individual symptoms between ASD and DCC, without examining the relationships among symptoms or their association with connectivity patterns. However, simply contrasting the presence or severity of individual symptoms remains relatively coarse, as many traitsâ€”such as social difficulties, anxiety, attentional alterations, or repetitive behaviorsâ€—may appear superficially similar across conditions. A more theoretically constraining approach consists in examining correlations, both between behavioral dimensions and between behavioral and neuroimaging measures. Robust behavioralâ€“behavioral associations are statistically less likely to arise incidentally and may reveal an underlying organizational principle linking symptoms into a coherent architecture. Likewise, behavioralâ€“neuraimaging correlations provide stronger mechanistic leverage by directly testing whether specific connectivity alterations map onto particular symptom configurations. Shared correlation patterns across conditions would suggest convergent integrative mechanisms, whereas divergent patterns would indicate distinct underlying architectures despite phenotypic overlap. Thus, moving beyond symptom comparison to correlation-based analyses offers a more discriminative and mechanistically informative framework for understanding similarities and differences between ASD and DCC.

Building on this rationale, the correlation architecture described in autism, encompassing both behavioralâ€“behavioral and behavioralâ€“neural associations, serves as a reference framework for the present investigation. Assessing whether these structured patterns are preserved or altered in DCC allows for a more discriminative comparison than symptom-level contrasts alone.

There is currently no direct study comparing ASD and DCC in terms of brain connectivity and its relationship to clinical symptoms. In autism research, numerous studies have identified significant associations between specific patterns of functional connectivity â€" including both hypo- and hyperconnectivity within and between large-scale networks â€" and symptom dimensions such as social difficulties, repetitive behaviors, and sensory hypersensitivity. However, these connectivityâ€“symptom correlations have not been systematically examined in DCC within a comparable framework. In the present study, we therefore aim to revisit the established correlations between brain connectivity and symptomatology in ASD and to investigate whether similar associations can also be observed in individuals with DCC, as a first step toward a structured comparison of connectivityâ€“symptom relationships across the two conditions.

Rather than reflecting a single core deficit, autism spectrum disorder (ASD) has been proposed to arise from a compound of partially dissociable neurocognitive differences that may co-occur to varying degrees across individuals (Booth et al., 2011). Each dimension has been associated with distinct but overlapping patterns of structural and functional connectivity alterations. Examining these dimensions separately allows more precise comparison with conditions such as developmental corpus callosum dysgenesis (DCC), in which interhemispheric integration is primarily affected.

Difficulties in social cognitionâ€"including theory of mind, emotion perception, and social inferenceâ€"have consistently been linked to reduced long-range functional connectivity within the default mode network (DMN), particularly between the medial prefrontal cortex (mPFC), posterior cingulate cortex/precuneus (PCC), and temporo-parietal junction (TPJ) (Assaf et al., 2010; Elton et al., 2015). The severity of social impairment correlates with decreased functional coherence within these midline DMN hubs and with reduced structural integrity of callosal fibers linking bilateral medial frontal and parietal associative cortices (Dimond et al., 2019), indicating that impaired interhemispheric integration constrains distributed social-cognitive processing.

Beyond cortico-cortical alterations, cerebellar contributions appear specifically tied to social dimensions. Ramos et al.Â (2019) reported altered functional connectivity between posterior cerebellar lobules (Crus I/II)â€"regions classically associated with higher-order cognitionâ€"and cortical areas implicated in social processing, including the mPFC and TPJ. Importantly, the strength of cerebello-cortical coupling in these circuits was associated with social symptom severity, suggesting that disrupted modulation from cerebellar associative territories may weaken predictive modeling and temporal coordination of social inference. This aligns with broader evidence of reduced connectivity between the cerebellum and supramodal associative corticesâ€"namely prefrontal, posterior parietal, and inferior/middle temporal regions that support multimodal abstraction and contextual integration (Khan et al., 2015). Given the cerebellumâ€™s role in fine-tuning distributed cortical activity via Purkinje-cell-mediated inhibitory regulation (Belmonte et al., 2004), reduced coupling with these integrative hubs may further destabilize large-scale coordination required for social cognition.

Communication impairmentsâ€"particularly pragmatic language and contextual comprehensionâ€"rely on partially overlapping but more lateralized networks. Effective pragmatic communication requires dynamic coordination between left inferior frontal gyrus (Brocaâ€™s area), left superior temporal gyrus, and right-hemisphere homologues involved in prosody and socio-emotional inference, placing strong demands on interhemispheric transfer. Altered connectivity within the executive control network (ECN), especially involving dorsolateral prefrontal cortex (dlPFC) and posterior parietal cortex, has been reported in ASD. Notably, in autistic individualsâ€"but not controlsâ€"ECN functional connectivity correlates with genu size of the corpus callosum (Booth et al., 2011), indicating a structureâ€“function relationship specifically linking callosal integrity to executive modulation of communication. Together, these findings suggest that communicative vulnerability is closely tied to disrupted bilateral associative integration and executive coordination.

Restricted and repetitive behavior

In contrast to social-communicative alterations, the restricted and repetitive behaviours (RRB) dimension of ASDâ€"including atypical sensory reactivityâ€"appears to rely more strongly on alterations in large-scale attentional and salience systems. Converging evidence implicates atypical interactions among the salience network (SN), executive control network (ECN), and dorsal attention network (DAN), together with thalamocortical, fronto-striatal, and cerebellar circuits. Broadly, the SN supports salience detection and network switching, the ECN mediates top-down control and flexibility, and the DAN sustains goal-directed orienting.

Importantly, the DSM-5 newly includes sensory hyper- or hyporeactivity within the RRB domain. Sensory hypersensitivity severity has been associated with inefficient attentional filtering, whereby reduced top-down modulation leads to amplification of incoming sensory signals (Gomes et al., 2008). Greater sensory hypersensitivity severity correlates with hyperconnectivity within the SNâ€”particularly involving the anterior insula and amygdalaâ€”and increased coupling between the SN and primary sensory cortices (Green et al., 2016). In parallel, thalamocortical hyperconnectivity, especially in auditory pathways, suggests reduced thalamic inhibition and diminished sensory gating (Baran et al., 2023). Together, these alterations indicate excessive propagation of salient sensory input and heightened arousal. Within this context, Liss et al.Â (2006) proposed that overselective attention may function as a compensatory mechanism: predictable and repetitive behaviours may reduce arousal by constraining attentional scope and stabilizing salience processing. Repetitive behaviours would thus not merely reflect inflexibility, but also serve a regulatory function in the face of sensory over-responsivity. Complementing this pattern, increased cerebellar connectivity with sensorimotor cortices and reduced coupling with supramodal associative regions (Khan et al., 2015), together with reports of cerebellar hypoplasia in ASD, suggest a shift toward low-level sensorimotor integration that may reinforce repetitive motor and cognitive routines.

Beyond this regulatory account, altered endogenous attentional control may further contribute to rigidity. The SN normally enables switching between internally oriented processing (e.g., default mode network activity) and externally oriented control and orienting systems, including the ECN and DAN. Disruption of this mechanism may reduce attentional flexibility. Altered ECN and DAN connectivity has been associated with impaired attention switching and cognitive inflexibility (Liss et al., 2006; Green et al., 2016). The monotropism account conceptualizes this pattern as a narrowing of attentional scope, in which cognitive resources become concentrated on a limited set of interests. Although the DAN can sustain strong engagement during focused tasks, its capacity to disengage and shift may be reduced. Finally, fronto-striatal circuitry differences involving the orbitofrontal cortex and caudate nucleus further support the contribution of habit-based loops to behavioural rigidity (Langen et al., 2011).

Notably, repetitive features appear less prominent in developmental corpus callosum dysgenesis (Badaruddin et al., 2007), indicating that disrupted interhemispheric integration alone may not be sufficient to generate the salience-driven sensory amplification and associated behavioural rigidity characteristic of ASD.

3.1 Core behavioral dimensions of autism

- defined behaviorally (contrary to DCC)
- various symptoms : â€¦.
    - Social interaction difficulties
    - Altered social and pragmatic communication
    - Restricted and repetitive behaviors
    - Cognitive rigidity
    - Atypical sensory processing
    - Attention : atypical regulation of attention rather than deficit
- Theory to explain this variety of symptoms ?
    - Executive function theory
    - Weak central coherence / monotropism -> matter of attention hypo/hyper
        - bad integration -> would result in low range of interest
    - social cognition deficit
    - Complexity deficit
    - ben-yosef2016 compares the last 3
        - CCL : diff when synchronization between different regions in required, but NOT ONLY, also cognitive overload in general
        - supports complexity account of ASD
    - overreactivity (leads to overselective attention and repetitive -> compensation) -> sensory processing could be the primary difference
    - attention makes the link between sensory and social profile

3.2 Mesure

- Use of dimensional rather than categorical approaches
- Social Responsiveness Scale â€“ Second Edition (SRS-2)

3.4 Connectivity models in autism

- Long-range hypoconnectivity
- Local hyperconnectivity
- Imbalance between global integration and local specialization

* Interhemispheric connectivity alterations, including the corpus callosum
* Limitations of unitary connectivity models in accounting for clinical heterogeneity

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4. Comparison between CCD and idiopathic autism

demopoulo 2015

The corpus callosum (CC) plays a central role in coordinating activity between the two hemispheres, enabling the integration of information across distant cortical regions. Callosal fibers, primarily excitatory axons from pyramidal neurons, terminate on contralateral GABAergic interneurons, providing transcallosal inhibition that regulates interhemispheric excitation and supports synchronized long-range communication. This inhibitory control is crucial: it prevents runaway local excitation and allows distant regions to communicate effectively, maintaining a balance between local and global connectivity. In autism, structural and functional studies indicate that the CC is altered: overall reductions in CC size are reported, particularly in the anterior body and rostral regions connecting frontal and parietal cortices, which are critical for working memory and higher-order integration (Piven et al., 1997). Functional studies show underconnectivity in fronto-parietal networks, correlated with the size of the genu (Just et al., 2006), and DTI meta-analyses reveal decreased fractional anisotropy throughout the CC, supporting reduced microstructural integrity and long-range signaling (Demopoulos et al., 2015). These deficits likely weaken interhemispheric inhibition and long-range connectivity, which, combined with local overconnectivity from excessive plasticity, may underlie the characteristic pattern of strong local processing but poor integration across distant regions observed in autism.

3.1 Phenotypic similarities/differences

* Social cognition difficulties
* Cognitive rigidity
* Elevated scores on dimensional measures (e.g., SRS-2)
* studies that directly compared it
    * lighter on almost every aspect ?
    * ARX ?

3.2 Comparison of the connectivity patterns

-> Would it be that different causes induced similar connectivity patterns and thus same behavioral patterns ? -> Are SRS2 scores linked between the 2 inderecty through connectivity patterns ?

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4. The missing link: brain connectivity and autistic traits

4.1 Current state of the literature
* Scarcity of studies directly linking
* structural connectivity (DTI)
* functional connectivity (fMRI)
* dimensional measures of autistic traits (SRS-2)

4.2 Methodological limitations
* Small sample sizes in CCD studies
* High phenotypic and neuroanatomical heterogeneity
* Predominantly descriptive approaches
* Limited fine-grained correlational analyses

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5. Integrative hypothesis and modeling

5.1 Main hypothesis
* Autistic traits observed in individuals with CCD may emerge from
* early disruption of interhemispheric integration
* combined with large-scale network reorganization
* leading to a behavioral phenotype partially overlapping with autism

5.2 Proposed conceptual model

* Structural alteration â†’ atypical connectivity â†’ atypical social cognition â†’ elevated SRS-2 scores

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6. Objectives of the present study

- To characterize brain connectivity in individuals with corpus callosum dysgenesis using
  - functional connectivity (fMRI)
  - structural connectivity (DTI)
- To examine relationships between connectivity measures and autistic traits
- To correlate connectivity patterns with SRS-2 scores
- To evaluate whether connectivity alterations in CCD resemble or differ from those described in idiopathic autism